

Brandon Bouley

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EDUCATION

Merrimack College, North Andover, MA
Bachelor of Science in Computer Science
Data Science Minor
Expected: May 2026 | GPA: 3.5

COURSES

Data Structures, Programming Languages, Algorithms, Discrete Mathematics, Databases, Object-Oriented Design, Operating Systems, Computer Graphics, AI, Network Security, Web Development, Computer Architecture, Mobile Application Development, Computer Science Capstone

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, SQL, HTML, CSS, Kotlin, Lua, C, R, ML, ProLog, Basic
Libraries and Tools: Three.js, Swing, SWT, React, Node.js, TensorFlow, PyTorch, SuperGradients, Pillow
Developer Tools: GitHub, Visual Studio Code, NetBeans, PyCharm, Jupyter Notebook, Android Studio

WORK EXPERIENCE

Merrimack College, N. Andover, MA

January 2026 - Present

Work With a Professor Directed Study

- Designing and implementing an intelligent work-schedule optimization web application tool integrating biomathematical models of sleep, circadian rhythms, and human performance to reduce fatigue and circadian misalignment.
- Solving a constrained optimization problem using techniques such as mixed-integer programming or constraint programming while satisfying operational, legal, and coverage requirements.
- Meeting weekly with a professor to report on work completed and future goals for the upcoming week.

CVS Pharmacy, Salem, NH

Shift Supervisor

September 2022 - Present

Store Associate

November 2021 - September 2022 (Promoted in 10 months)

- Recognized for leadership in customer service, daily operations, and team development.
- Supervise daily store operations by delegating tasks to a team of up to 5 employees, monitoring performance, and ensuring completion of shift responsibilities.
- Resolve an average of 1-5 customer complaints per shift by providing prompt assistance, troubleshooting problems, and delivering effective solutions to resolve issues.
- Manage \$5-\$10K in daily transactions with <1% error rate, demonstrating accuracy and accountability.
- Train an average of 3 new employees every month, improving team performance and onboarding efficiency by 25%.

Town of Andover Health Department, Andover, MA

Compliance Checker

March 2024 - May 2025

- Conducted an average of 10 random compliance checks per shift at retailers in northeastern Massachusetts.
- Attempted controlled purchases to identify if retailers are selling tobacco products to underage individuals.
- Documented findings and reported results to my compliance officer.
- Supported community health initiatives aimed at reducing youth access to tobacco and nicotine products.

PROJECTS

Web App - Oral History Translation Tool | *Python, XML, JavaScript* *September 2025 - December 2025*

- Created a real-time translation web app that translates audio and video files into English, French, Portuguese, and Spanish.
- Built-in edit mode for creating changes to both the original text and the translated version.
- Utilized Google Cloud technologies, such as buckets for data storage and OAuth for user authentication.
- Collaborated with a team in an agile environment, participating in sprint planning and iterative feature delivery.

Web Development - Personal Portfolio | *HTML, CSS, JavaScript* *September 2025*

- Develop a clean, simple-to-use webpage to present technical projects and coursework, emphasizing usability and front-end structure.
- Implemented basic UI elements using **HTML** and **CSS**, while also creating interactive elements using **JavaScript**.

Machine Learning - Parking Spot Detector | *Python* *April 2025 - May 2025*

- Built a real-time AI system that detects and classifies parking spots as occupied or vacant.
- Built and curated a large dataset, splitting it into training, validation, and testing sets for robust model performance.
- Implemented and trained deep learning models using **Python** libraries, including **TensorFlow**, **PyTorch**, **SuperGradients**, and **Pillow** to optimize training and image classification accuracy.

3D Video Game | *JavaScript & HTML* *December 2024*

- Collaborated with a team of three people to build a fully functional 3D version of the game “Plants vs. Zombies” that allows players to place plants on a lawn that can defend against attacking zombies.
- Imported pre-made 3D models of plants and zombies using **GLTF**, and used **JavaScript** to animate their movements.
- Learned how to use **HTML** for various GUI containers and elements, including a toolbar where players could select different tools and plants.

Systems Programming - Deadlock Detector | *C*

- Developed a command-line deadlock detection tool to analyze potential deadlock situations occurring from multithreading conflicts.
- The tool recommended fixes in the user’s code to potentially fix any situations where a deadlock is possible.
- Applied operating systems concepts, including synchronization primitives, memory management, and modular code organization to improve correctness and maintainability.

Custom Programming Language | *Java* *September 2023- December 2023*

- Collaborated with a peer to build and design a functional programming language within **Java** that featured higher-order functions, recursion, and immutable data structures.
- Implemented the building blocks and the internal workings of a programming language, such as syntax, data types, variables, control structures, and parsing.
- Utilized **GitHub** for version and source control, managing changes through branches, commits, and pull requests.

Robotics Programming | *Python & Basic* *Fall 2023 - Fall 2024*

- Collaborated with a team to assemble, program, and test robots for competitive events.
- Developed and debugged code using the VEX Robotics extension in **Visual Studio Code**, strengthening hands-on experience.
- Programmed both autonomous and driver-controlled functionalities in **Python** and **Basic** languages.